Data Journalism

Session 3: How people understand numbers and visuals

Simon Munzert
Hertie School | GRAD-E1493

DJ critique

Discuss

1. **"So what":** What are the key take-away? And what role does the data play for establishing it?
2. **Sources & access:** What data did they use and how did they get it? Could you reproduce their analysis? What's missing?
3. **Design and storytelling:** Does their scrollytelling approach work for you? Are there flaws in visual and narrative design that you would want to be fixed? How?

Source Wired 2023

Today

1. Data and visual literacy in the public
2. Communicating data: evidence on what works
3. Communicating uncertainty
4. Discussion

Your readers are not statisticians

The core tension

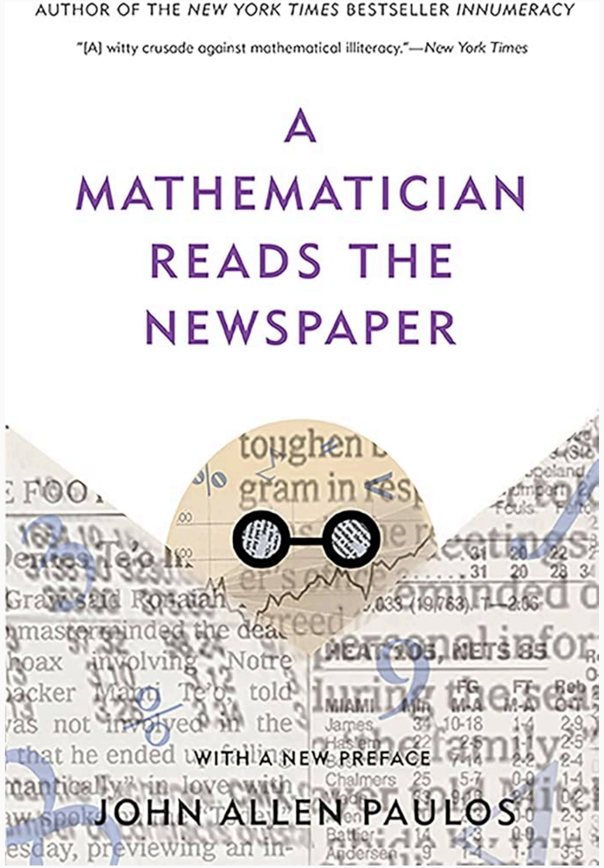
You work hard to get the numbers right — but does your audience get them?

Data journalism often assumes a **numerically literate reader**. That assumption is likely wrong.

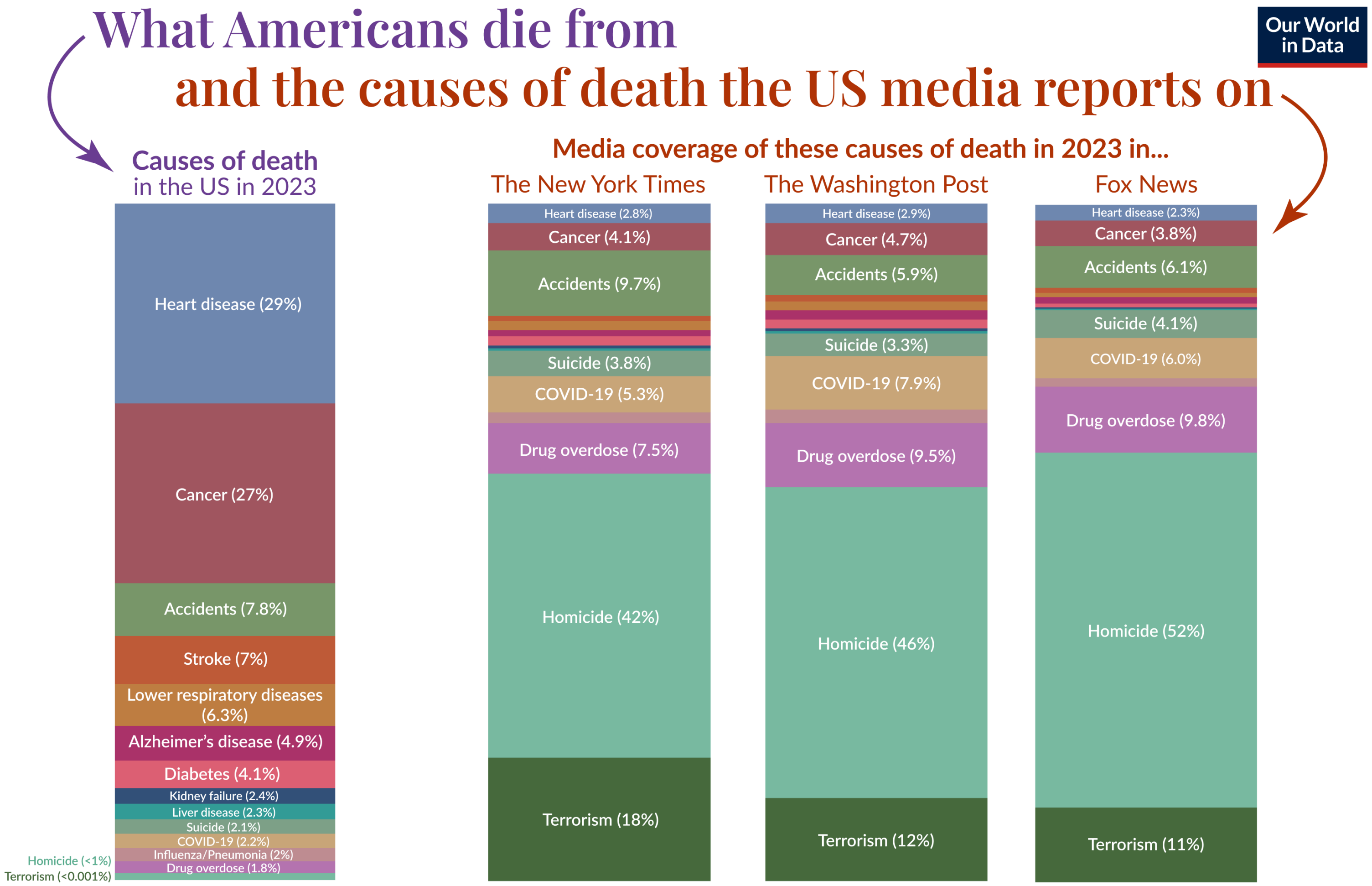
What's at stake

- Misread statistics → misformed opinions on policy, health, elections
- Misleading visuals → distorted perception of risk or scale
- Overconfident forecasts → eroded trust in media and science

This session is about the **empirical evidence** on how people actually process data and visuals — and what happens if things go wrong.



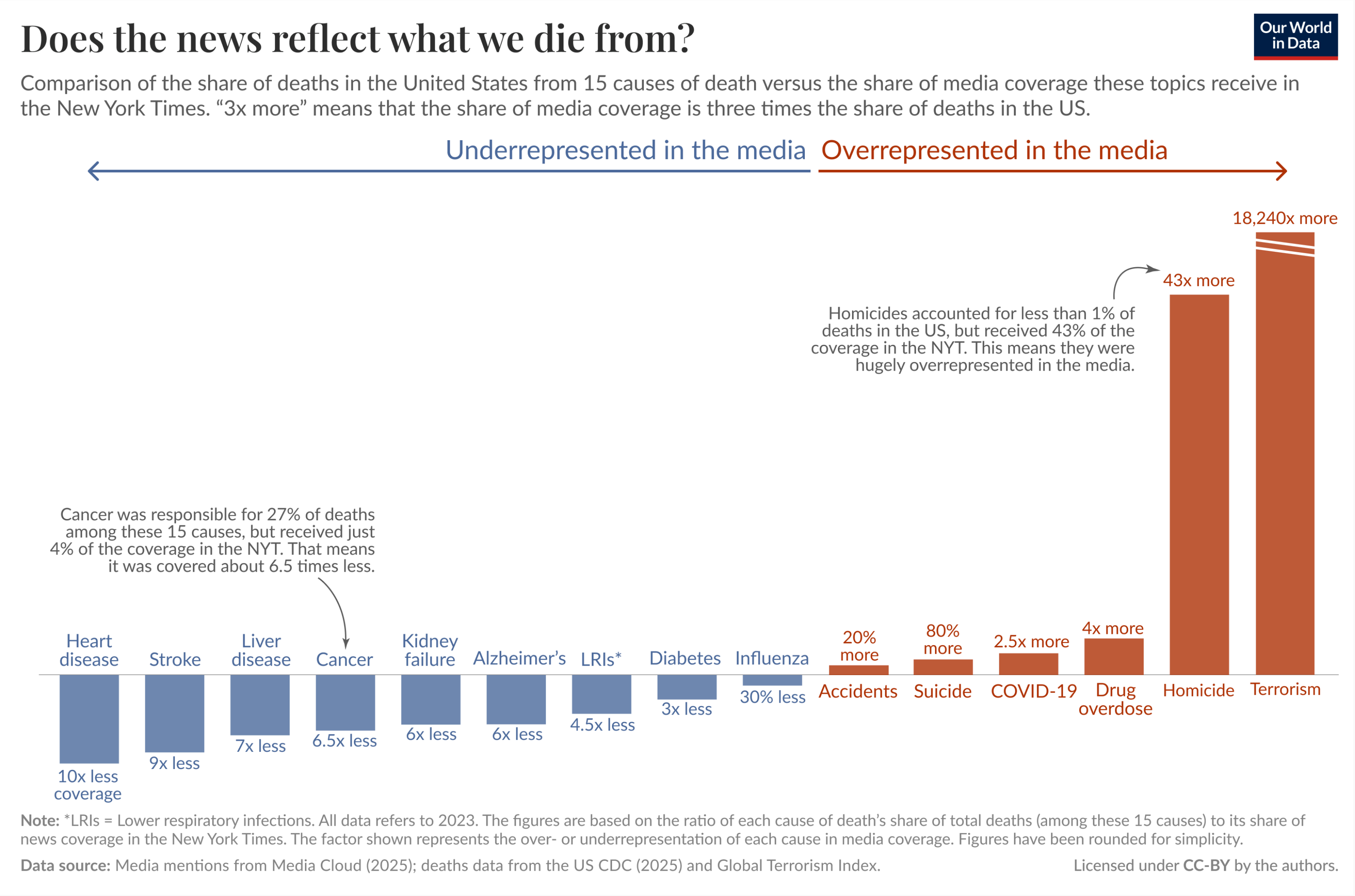
Also, journalists are not statisticians (but people with incentives)



Note: Based on the share of causes of death in the US and the share of mentions for each of the causes in the New York Times, the Washington Post, and Fox News. All values are normalized to 100%, so the shares are relative to all deaths caused by the 12 most common causes + drug overdoses, homicides, and terrorism. These causes account for more than 75% of deaths in the US. A "media mention" is a published article in one of the outlets which mentions the cause (e.g., "influenza") or related keywords (e.g., "flu") at least twice.

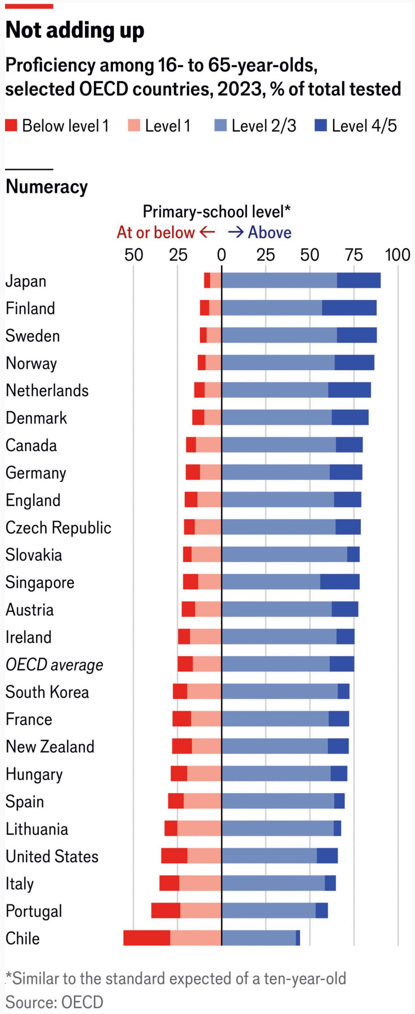
Data sources: Media mentions from Media Cloud (2025); deaths data from the US CDC (2025) and Global Terrorism Index.

Also, journalists are not statisticians (but people with incentives)



Data and visual literacy in the public

How numerate is the public?



Source The Economist, Dec 2024

OECD Programme for the International Assessment of Adult Competencies (PIAAC)

- ~160,000 adults across 33 countries tested on literacy, numeracy, problem-solving
- **~50% of adults in OECD countries score at or below Level 2** (out of 5) on numeracy
- Level 2 = can handle simple arithmetic with whole numbers; struggles with percentages, fractions, probability

What this means for our audience

- Percentages and ratios already challenge many readers
- "1 in 100,000" vs "0.001%" — same, but not perceived as same
- Denominator neglect: absolute numbers feel bigger than rates

The Berlin Numeracy Test

Judgment and Decision Making, Vol. 7, No. 1, January 2012, pp. 25–47

Measuring Risk Literacy: The Berlin Numeracy Test

Edward T. Cokely*† Mirta Galesic,† Eric Schulz†‡ Saima Ghazal§
Rocio Garcia-Retamero†¶

Abstract

We introduce the Berlin Numeracy Test, a new psychometrically sound instrument that quickly assesses statistical numeracy and risk literacy. We present 21 studies (*n*=5336) showing robust psychometric discriminability across 15 countries (e.g., Germany, Pakistan, Japan, USA) and diverse samples (e.g., medical professionals, general populations, Mechanical Turk web panels). Analyses demonstrate desirable patterns of convergent validity (e.g., numeracy, general cognitive abilities), discriminant validity (e.g., personality, motivation), and criterion validity (e.g., numerical and non-numerical questions about risk). The Berlin Numeracy Test was found to be the strongest predictor of comprehension of everyday risks (e.g., evaluating claims about products and treatments; interpreting forecasts), doubling the predictive power of other numeracy instruments and accounting for unique variance beyond other cognitive tests (e.g., cognitive reflection, working memory, intelligence). The Berlin Numeracy Test typically takes about three minutes to complete and is available in multiple languages and formats, including a computer adaptive test that automatically scores and reports data to researchers (www.riskliteracy.org). The online forum also provides interactive content for public outreach and education, and offers a recommendation system for test format selection. Discussion centers on construct validity of numeracy for risk literacy, underlying cognitive mechanisms, and applications in adaptive decision support.

Keywords: risk literacy, statistical numeracy, individual differences, cognitive abilities, quantitative reasoning, decision making, risky choice, adaptive testing, Mechanical Turk.

Appendix III: Berlin Numeracy Test traditional paper and pencil format

Instructions: Please answer the questions below. Do not use a calculator but feel free to use the space available for notes (i.e., scratch paper).

1. Imagine we are throwing a five-sided die 50 times. On average, out of these 50 throws how many times would this five-sided die show an odd number (1, 3 or 5)? ________ out of 50 throws.
2. Out of 1,000 people in a small town 500 are members of a choir. Out of these 500 members in the choir 100 are men. Out of the 500 inhabitants that are not in the choir 300 are men. What is the probability that a randomly drawn man is a member of the choir? (please indicate the probability in percent). ___________ %
3. Imagine we are throwing a loaded die (6 sides). The probability that the die shows a 6 is twice as high as the probability of each of the other numbers. On average, out of these 70 throws, how many times would the die show the number 6? ___________ out of 70 throws.
4. In a forest 20% of mushrooms are red, 50% brown and 30% white. A red mushroom is poisonous with a probability of 20%. A mushroom that is not red is poisonous with a probability of 5%. What is the probability that a poisonous mushroom in the forest is red? _____________%

Related results

- Original paper: mean score ~1.6/4 across university samples; score predicts investment decisions, medical risk understanding
- Framing effects are larger for low-numeracy readers (Peters et. al 2006; Zhu and Feldman 2023)

Test yourself: riskliteracy.org

The INSPIRE inventory (work in progress)

Measuring scientific evidence consumption literacy for public policy: Development and validation of the INSPIRE inventory

Sebastian Ramirez-Ruiz (Hertie School)*
Simon Munzert (Hertie School)

Abstract. Being able to make sense of scientific evidence is a critical component in shaping effective public policy, enabling both policymakers and the public to make informed decisions on complex issues. However, existing measures of this skillset may not fully address the specific methodological and applied demands of public policy contexts. We develop the *Inventory for Numeracy, Statistics, and Policy-oriented Inference and REasoning* (INSPIRE), a comprehensive measurement instrument to assess competence in scientific evidence consumption in relation to public policy. The instrument integrates knowledge of statistical reasoning, data and visualization literacy, causal reasoning and inference, and the scientific method, alongside the ability to critically evaluate scientific information pertinent to policy debates. Using Item Response Theory (IRT), we systematically assess an initial item pool, resulting in an inventory of 30 items. We assessed the inventory's psychometric and substantive validity across three samples: a general population, policy students and professionals in a data science training program, and participants in a pre-election forecasting study. This allowed us to examine its performance across different groups and contexts. Results indicate good internal consistency and evidence of construct, criterion, and predictive validity. We close with considerations for potential use of the item bank in research and applied settings.

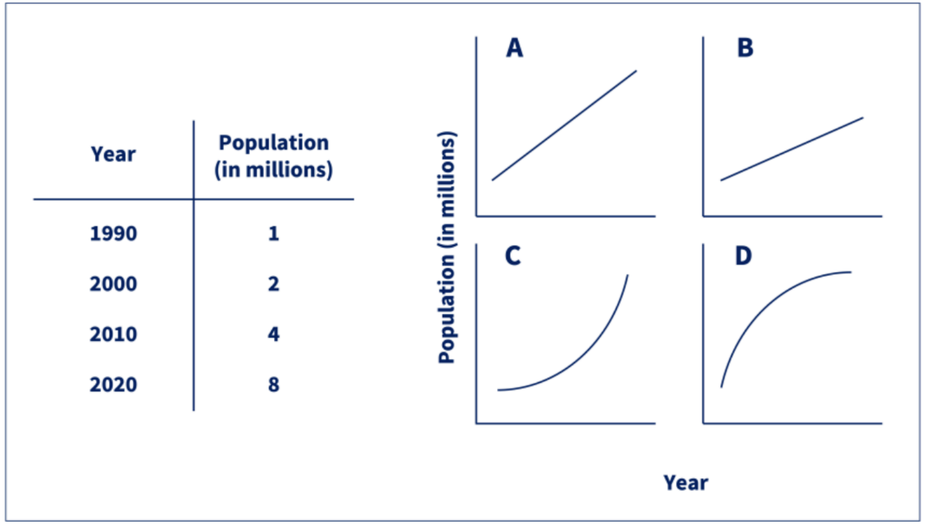
Domain and sub-domain	Item code	Item label	2PL Disc (a)	2PL Diff (b)	Percent correct	Median time (s)	N
Scientific Literacy							
Basic science	SEL04	Research credibility	1.73	-2.99	98.0%	14	175
Basic science	SEL03	Quality of evidence	1.91	-2.37	95.0%	18	170
Source reliability	SEL07	Trustworthy data	1.54	-2.39	94.0%	13	190
Source reliability	SEL08	Trustworthy sources	1.42	-2.49	94.0%	24	172
Scientific practice	SEL06	Scientific practice 2	1.65	-1.92	90.0%	21	176
Basic science	SEL01	Scientific hypothesis	1.61	-1.74	88.0%	10	189
Scientific practice	SEL05	Scientific practice 1	0.89	-2.44	85.0%	22	175
Basic science	SEL02	Scientific consensus	1.06	-0.03	50.0%	9	176
Statistical Literacy							
Machine learning models	SML05	ML and perpetuated bias	1.96	-2.68	97.0%	8	220
Machine learning models	SML07	ML application	1.04	-2.52	91.0%	10	210
Base rate	SML03	Base rate 1	1.40	-1.59	88.0%	39	160
Machine learning models	SML04	ML accuracy	1.27	-1.69	84.0%	9	234
Center and spread	SML01	Center and spread	0.82	-1.66	74.0%	36	163
Center and spread	SML02	Sample sizes and uncertainty	0.95	-0.66	65.0%	36	162
Machine learning models	SML06	ML and flawed predictions	0.82	-0.45	60.0%	29	193
Data Literacy							
Visual information	DVL06	Visual information 2	1.32	-2.97	95.0%	24	261
Numeracy	DVL01	Complementary probability 1	1.47	-1.97	90.0%	11	174
Visual information	DVL05	Visual information 1	0.86	-1.87	81.0%	48	241
Visual representation	DVL04	Visual data representation 1	0.99	-0.25	57.0%	33	235
Numeracy	DVL02	Complementary probability 2	1.65	-0.01	50.0%	31	182
Numeracy	DVL03	Percentages	0.93	0.23	48.0%	42	181
Problematic visuals	DVL07	Flawed visuals 1	1.14	0.50	41.0%	58	253
Problematic visuals	DVL08	Flawed visuals 2	0.85	0.89	37.0%	54	245
Causal Reasoning							
Experiments	CR04	Randomized control trials	0.80	-4.38	96.0%	10	160
Causal reasoning	CR05	Confounder	1.09	-1.90	85.0%	24	196
Valid causal conclusions	CR07	Study conclusions 1	2.09	-1.12	82.0%	26	181
Correlation	CR03	Learning from correlation	0.86	-0.97	69.0%	27	192
Causal policy	CR06	Causal effect of a policy	1.08	-0.65	64.0%	27	193
Correlation	CR02	Correlation	0.82	-0.49	60.0%	20	197
Correlation	CR01	Correlation vs. Causation	1.04	0.22	46.0%	10	168

Table 2: Overview of items in the INSPIRE inventory. This table presents item-level estimates from a two-parameter logistic (2PL) IRT model fitted to responses from N = 470 Prolific participants. For each item, the discrimination parameter (a) and difficulty parameter (b) are reported, alongside the sample share of correct answers and the median response time in seconds (measured as the time from item display to answer submission). A full list of items (including those that were excluded) and additional performance diagnostics are available in Online Appendix A.

The INSPIRE inventory (work in progress)

You have been keeping track of the population changes in a specific city over time. The table below shows your rough count of this population.

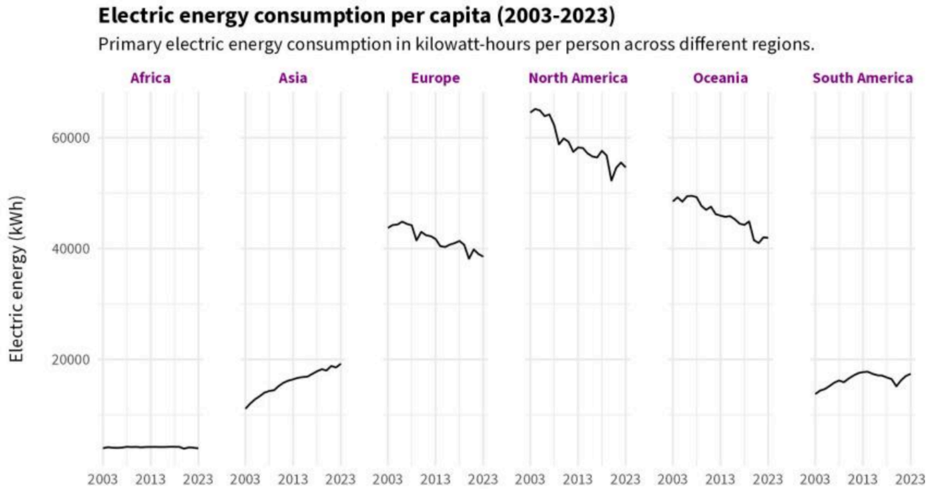
Which graph provides the **most suitable representation** of your data?



Year	Population (in millions)
1990	1
2000	2
2010	4
2020	8

- ◯ A
- ◯ B
- ◯ C
- ◯ D

Examine the following graph carefully. What can be learned from the graph?



- ◯ By 2050, Asia will surpass European electric energy consumption per capita levels.
- ◯ There is a decrease in electric energy consumption across the globe.
- ◯ North America is the highest per capita electric energy consumer across the regions.
- ◯ The rise in renewable energy sources resulted in a decrease of electric energy consumption in Europe, North America, and Oceania.

Brainstorming exercise (10 min)

The brief

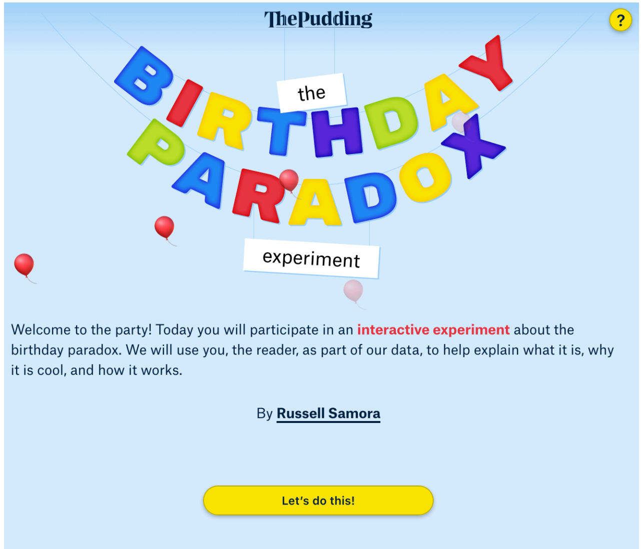
You're pitching to an editor at a major outlet. Turn the evidence from this block into a story.

Discuss in pairs and pick one angle

- 📰 Investigation
- 🎯 Interactive explainer
- 📊 Visualization critique
- Something else!

Report back (5 min)

- What's your pitch in one sentence?
- What would make it *not* just a meta-story about journalism?



Inspiration

- The Pudding: Birthday Paradox
- NYT: What's Going On in This Graph?
- NYT: You Draw it: How Family Income Predicts Children's College Chances
- Graphs.World

Communicating data: evidence on what works

Readings

JOURNALISM STUDIES
2021, VOL. 22, NO. 5, 575–594
https://doi.org/10.1080/1461670X.2021.1889398

Routledge
Taylor & Francis Group

Check for updates

Credibility and Enjoyment through Data? Effects of Statistical Information and Data Visualizations on Message Credibility and Reading Experience

Elena Link [a], Jakob Henke[b] and Wiebke Möhring[b]

[a]Department of Journalism and Communication Research, HMTM Hannover, Hannover, Germany; [b]Institute for Journalism, TU Dortmund, Dortmund, Germany

ABSTRACT
News organizations, journalists, and scholars are searching for ways to increase the perceived message credibility of the stories they produce. With the aim to evaluate the potentials of data-based storytelling formats, we build on message credibility research and investigate how features of message content such as the use of different presentation formats of statistical information and data visualizations affect news users' message credibility judgements as well as their reading experience. We conducted two online experiments to detect the influences of percentages (e.g., 33%) and verbalized ratios (e.g., one out of three) as well as static and interactive data visualizations. Results of both studies indicate that the format of statistical information and the interactivity of data visualizations do not affect message credibility and reading experience. However, the perceived interactivity of the graphic has an impact on the reading experience. One conclusion is that the extra work involved in creating interactive graphics does not seem to pay off directly in terms of message credibility and improving the reading experience.

KEYWORDS
Message credibility; reading experience; data journalism; statistical information; data visualization; experimental study

Swaying the Public? Impacts of Election Forecast Visualizations on Emotion, Trust, and Intention in the 2022 U.S. Midterms

Fumeng Yang, Mandi Cai, Chloe Mortenson, Hoda Fakhari, Ayse D. Lokmanoglu, Jessica Hullman, Steven Franconeri, Nicholas Diakopoulos, Erik C. Nisbet, Matthew Kay

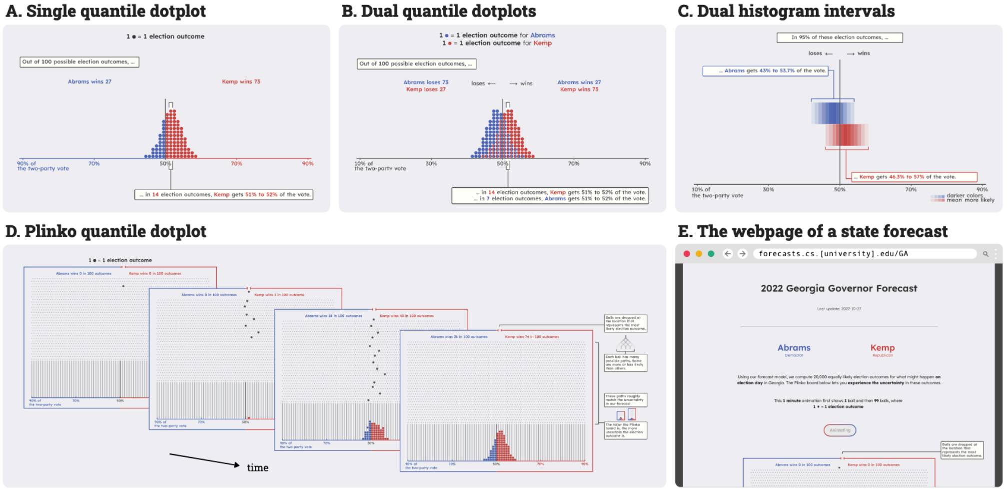


Fig. 1: **The four election forecast visualizations** in our longitudinal study: **A. Single quantile dotplot** (1-Dotplot), **B. Dual quantile dotplots** (2-Dotplot), **C. Dual histogram intervals** (2-Interval), and **D. Plinko quantile dotplot** (Plinko). In the 2022 U.S. midterm elections, we frequently updated this website to include the newest forecasts. Here all four visualizations show the forecast for Georgia on Oct. 27, 2022, which predicted a 73% probability of the Republican candidate Brian P. Kemp winning the governorship.

Abstract—We conducted a longitudinal study during the 2022 U.S. midterm elections, investigating the real-world impacts of uncertainty visualizations. Using our forecast model of the governor elections in 33 states, we created a website and deployed four uncertainty visualizations for the election forecasts: single quantile dotplot (1-Dotplot), dual quantile dotplots (2-Dotplot), dual histogram intervals (2-Interval), and Plinko quantile dotplot (Plinko), an animated design with a physical and probabilistic analogy. Our online experiment ran from Oct. 18, 2022, to Nov. 23, 2022, involving 1,327 participants from 15 states. We use Bayesian multilevel modeling and post-stratification to produce demographically-representative estimates of people's emotions, trust in forecasts, and political participation intention. We find that election forecast visualizations can heighten emotions, increase trust, and slightly affect people's intentions to participate in elections. 2-Interval shows the strongest effects across all measures; 1-Dotplot increases trust the most after elections. Both visualizations create emotional and trust gaps between different partisan identities, especially when a Republican candidate is predicted to win. Our qualitative analysis uncovers the complex political and social contexts of election forecast visualizations, showcasing that visualizations may provoke polarization. This intriguing interplay between visualization types, partisanship, and trust exemplifies the fundamental challenge of disentangling visualization from its context, underscoring a need for deeper investigation into the real-world impacts of visualizations. Our preprint and supplements are available at https://doi.org/osf.io/ajq8f.

Group discussion (15 min)

In groups of 2, discuss the following questions for one of the papers.

1. What are the key treatments studied in the paper, what are the key outcomes?
2. How is their research design set up to identify causal effects? Do you see strengths and weaknesses?
3. What are the main findings? Do they surprise you? Do they align with your intuition or experience?

Method

To answer our research questions, we conducted two online experiments with identical 2 × 2 between-subject-designs. In both experiments, the type of reporting statistical information (percentage versus verbalized ratios; cf. RQ1 and 4) and the visualization of statistical information (static versus interactive; cf. RQ2 and 5) in a news article were manipulated. In study 1 the article focused the tax burden in Europe (henceforth Tax). The article of study 2 discussed German crime statistics (henceforth Crime).

We collected data via a combined recruiting process using a German online survey panel (SoSci-Panel; Leiner 2016). Participants for both studies received the same link to the questionnaire and were then randomly assigned to either the Tax or Crime study. This leads to two separate and distinct samples. In the following, we will describe the procedure and descriptive results for both studies at the same time.

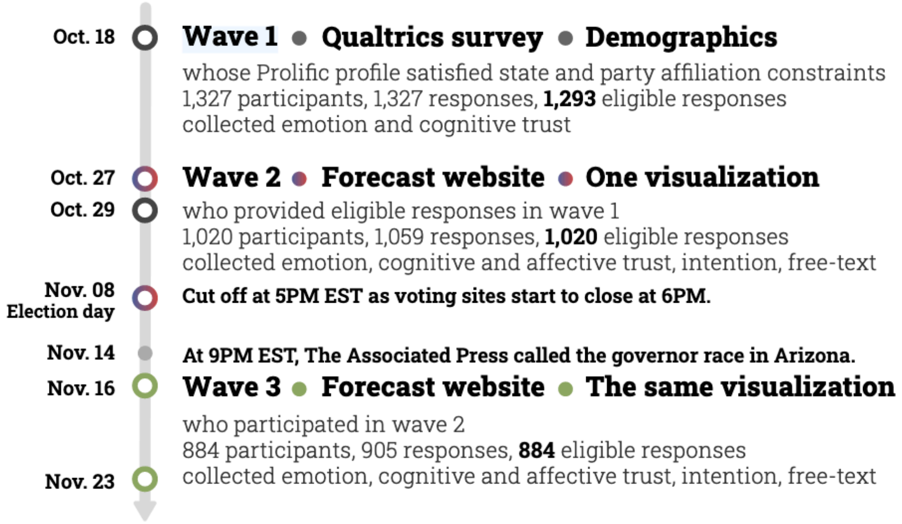


Fig. 5: **The design of the three-wave longitudinal study**, running from Oct. 18, 2022 to Nov. 23, 2022.

More experimental evidence on format effects

How the presentation of election polls shapes public trust and understanding*

Elias Koch[1,2], Linus Hagemann[1], Simon Munzert[1], Hannah Rajski[3,4], Thomas Gschwend[3,4], Cornelius Erfort[5], and Lukas F. Stoetzer[5]

[1]Data Science Lab, Hertie School Berlin
[2]Research Training Group DYNAMICS
[3]University of Mannheim
[4]Mannheim Centre for European Social Research (MZES)
[5]University of Witten/Herdecke

Working Paper

Pre-election polls shape how voters understand electoral competition, yet news outlets vary widely in how they present them. To assess how design and content choices in poll reporting influence audience perceptions, we fielded a large visual conjoint experiment during the 2025 German federal election (N = 23,695), including subsamples of voters (N = 23,444), journalists (N = 108), and parliamentary candidates (N = 143). Participants were presented with one of 160 versions of the same poll, randomly varying reporting formats and technical elements such as reference values, uncertainty information, and sampling details. Across all comparisons, any visualisation — bar charts, dot plots, or line graphs — were rated as substantially more attractive and more trustworthy than text-only formats, which still dominate real-world poll reporting, especially among less statistically literate participants. The effects of technical features were more mixed: some increased trust, others reduced perceived clarity, and few altered expectations about election outcomes. The findings demonstrate that the communication of polls meaningfully shapes their reception and provide evidence-based guidance for designing clearer and more trustworthy reporting of public opinion signals.

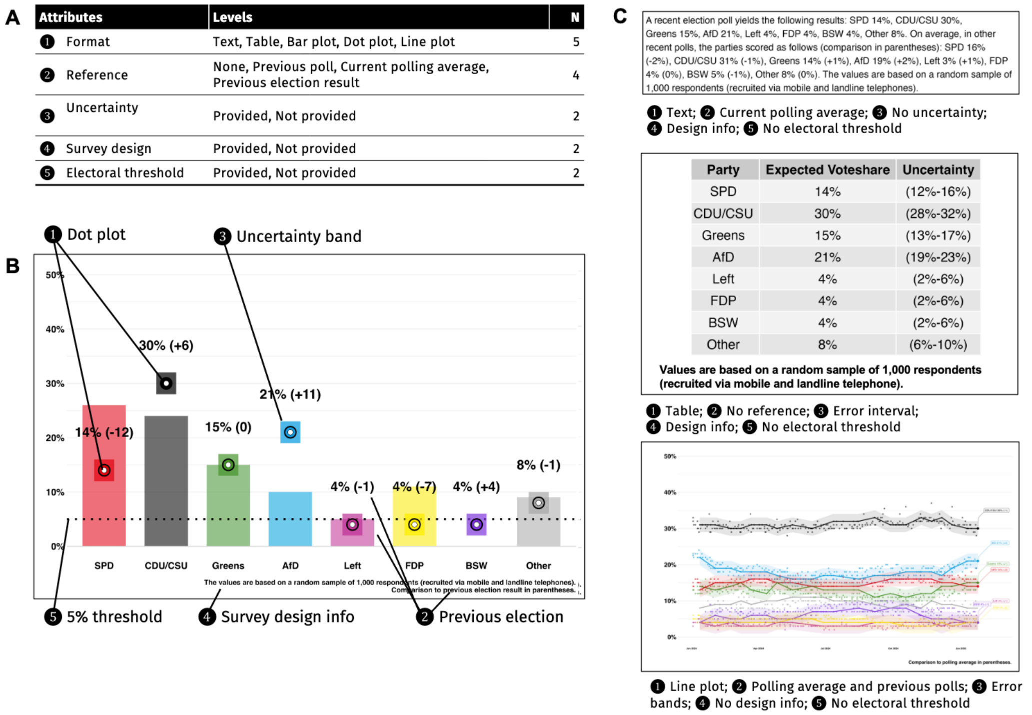


Figure 1: **Randomization setup of poll vignettes.** **(A)** Vignette attributes and levels, which were randomly combined into poll vignettes, resulting in $N_{\mathrm{msg}} = 160$ unique design variations. **(B)** Illustration of poll vignette: A dot plot enriched with additional information; **(C)** Further examples of vignettes with different attribute combinations. See SI A for an overview of the vignette construction and additional examples. All vignettes were presented in German; English translations above are provided solely for illustrative purposes.

More experimental evidence on format effects

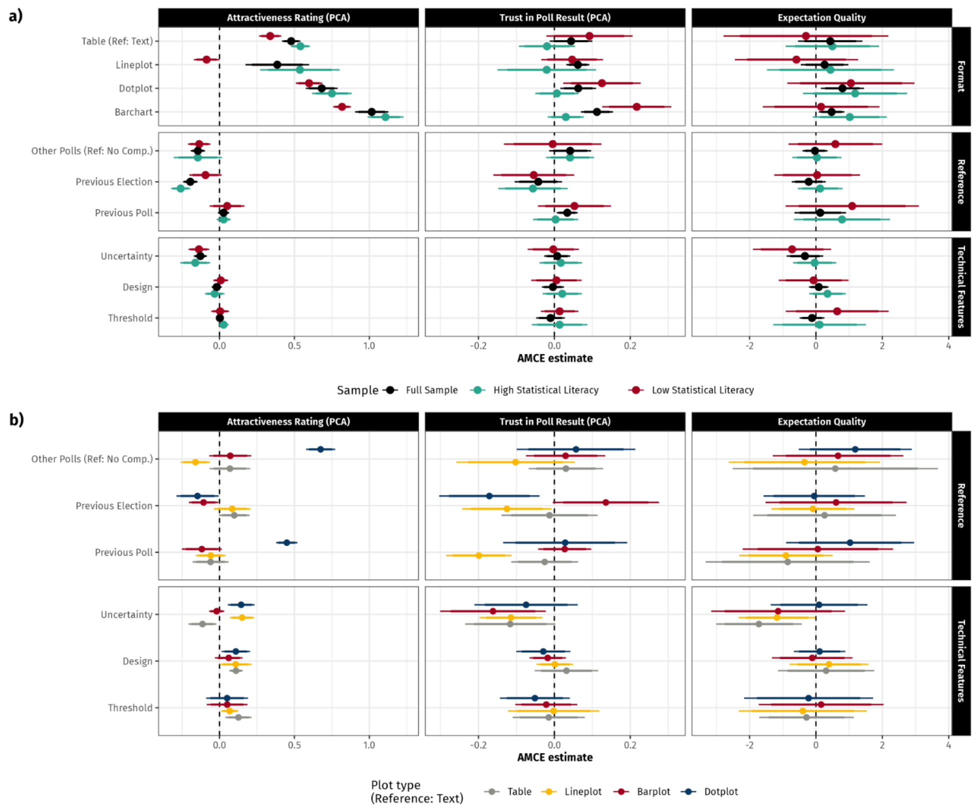


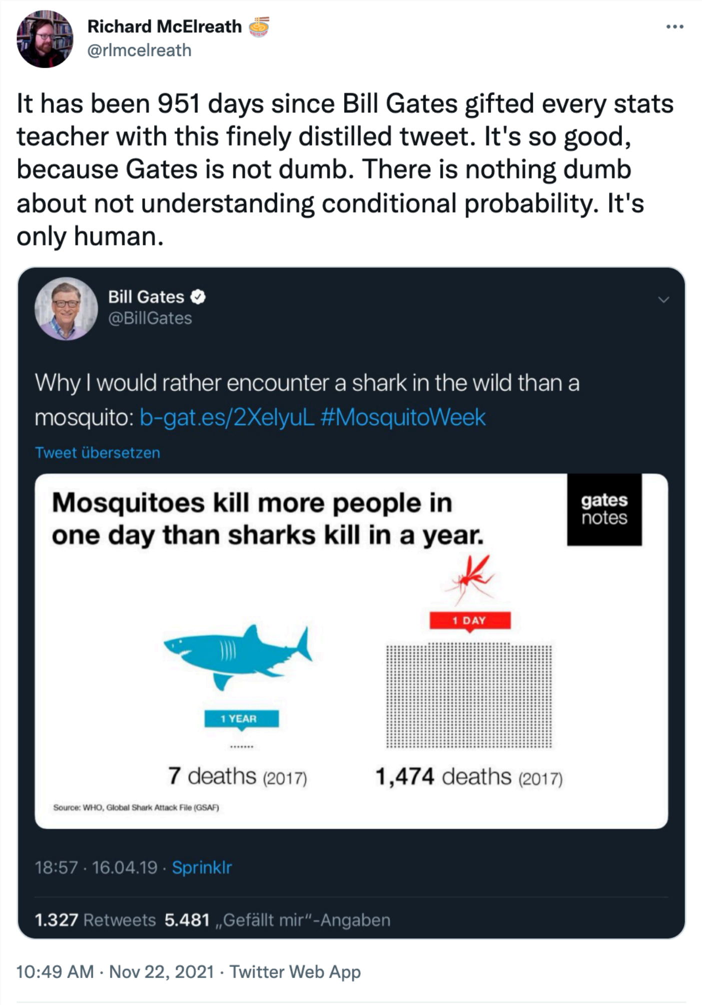
Figure 2: **Coefficient plots for both core analyses. Panel a)** presents baseline AMCE estimates for the different formats, reference values, and technical features with reference values in parentheses for non-dichotomous variations in dark blue. It further distinguishes between above (green) and below (red) median statistically literate respondents in the sample. **Panel b)** presents heterogeneous baseline effects of the inclusion of reference values and technical features in each visualisation format, compared to textual representations as a reference category. 95% (narrow) and 90% (wide) confidence intervals reported.

Communicating uncertainty

Communicating uncertainty

The difficulty of communicating uncertainty

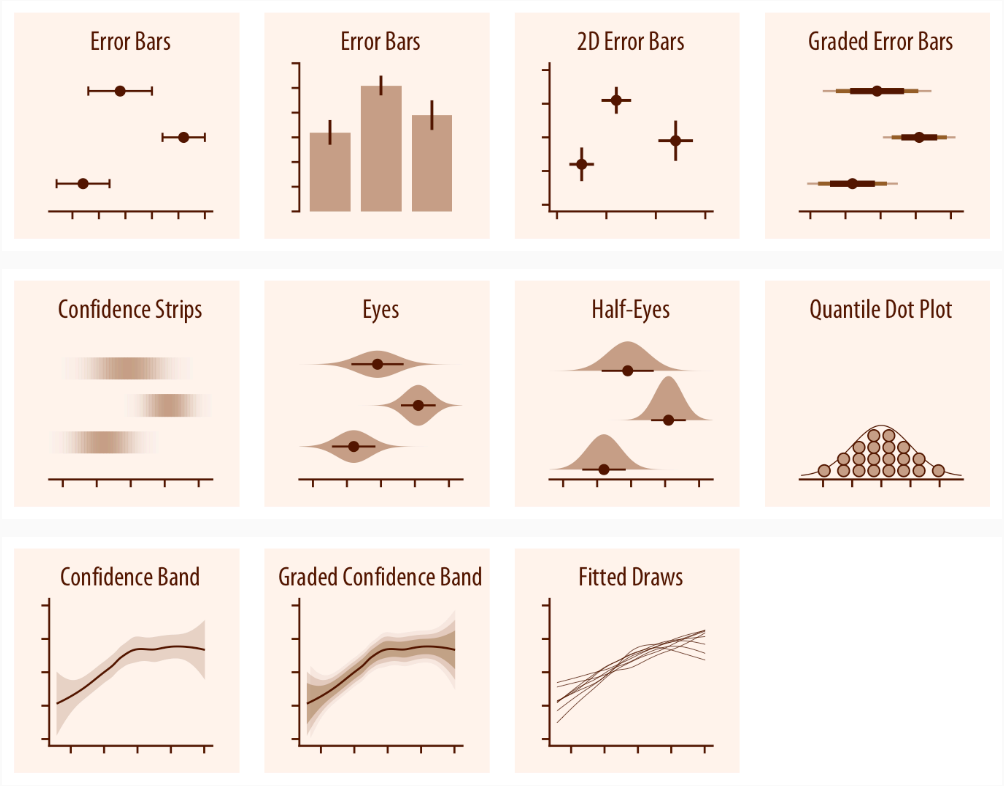
- The concept is complex. Not all people think in probabilistic terms.
- Many humans are bad at understanding (conditional and unconditional) probabilities.
- Adding information about uncertainty might distract, confuse, and undermine trust.



Credit Richard McElreath

Communicating uncertainty (cont.)

Visualizing uncertainty

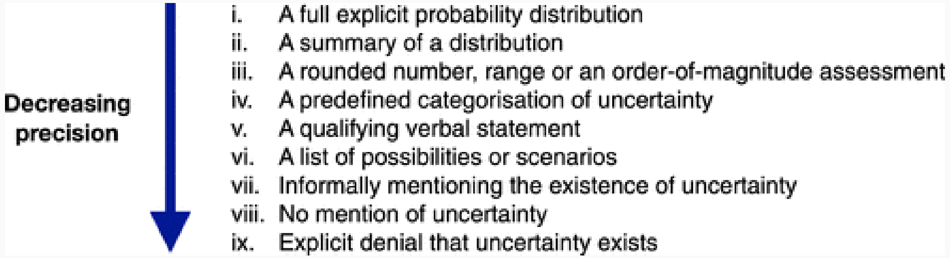


`Source` Claus Wilke

Uncertainty by numbers

```
> broom::tidy(model_out, conf.int = TRUE, conf.level = 0.95)
# A tibble: 4 × 7
  term        estimate std.error statistic   p.value conf.low conf.high
  <chr>          <dbl>     <dbl>     <dbl>     <dbl>    <dbl>     <dbl>
1 (Intercept) 13.4      0.175         76.7 0           13.1      13.8
2 distance    -0.00405  0.000110     -36.9 5.53e-297  -0.00426  -0.00383
3 originJFK   -2.70     0.189        -14.3 1.46e- 46  -3.07     -2.33
4 originLGA   -4.46     0.194        -23.0 3.04e-117  -4.84     -4.08
```

Strategies by precision



`Credit` van der Bles et al. 2019

Probabilities: confusing vote share with p(win)

Projecting Confidence: How the Probabilistic Horse Race Confuses and Demobilizes the Public

Sean Jeremy Westwood, Dartmouth College
Solomon Messing, Acronym
Yphtach Lelkes, University of Pennsylvania

Recent years have seen a dramatic change in horse-race coverage of elections in the United States—shifting focus from late-breaking poll numbers to sophisticated meta-analytic forecasts that emphasize candidates' chance of victory. Could this shift in the political information environment affect election outcomes? We use experiments to show that forecasting increases certainty about an election's outcome, confuses many, and decreases turnout. Furthermore, we show that election forecasting has become prominent in the media, particularly in outlets with liberal audiences, and show that such coverage tends to more strongly affect the candidate who is ahead—raising questions about whether they contributed to Trump's victory over Clinton in 2016. We bring empirical evidence to this question, using American National Election Studies data to show that Democrats and Independents expressed unusual confidence in a decisive 2016 election outcome—and that the same measure of confidence is associated with lower reported turnout.

I don't know how we'll ever calculate how many people thought it was in the bag, because the percentages kept being thrown at people—"Oh, she has an 88% chance to win!"
—Hillary Clinton quoted in Traister (2017)

`Source` Westwood et al. 2020

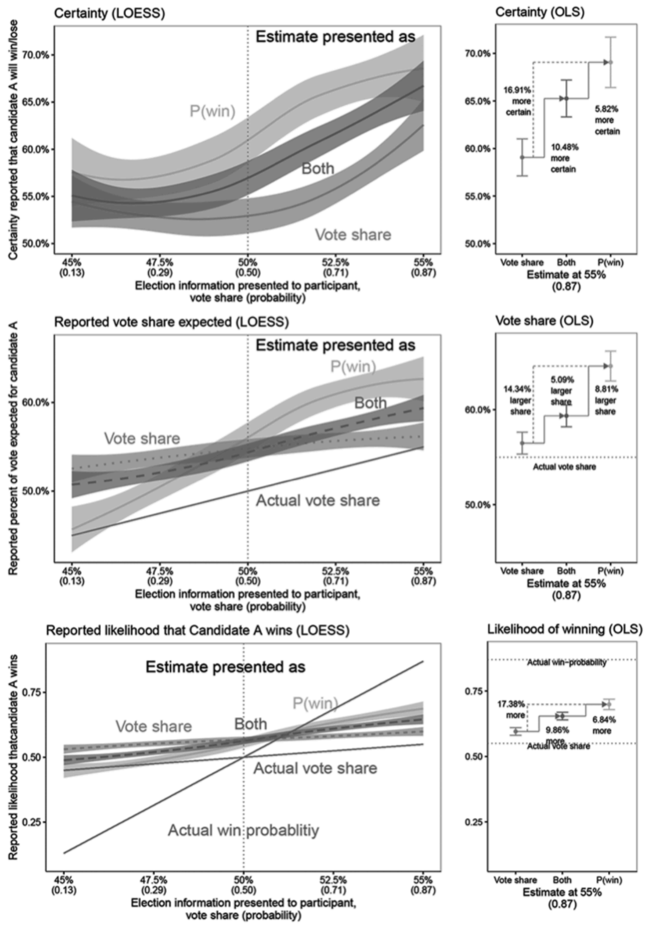


Figure 5. Effects of probabilistic forecasts on perceptions of an election. Probabilistic forecasts create the impression that the leading candidate will win more decisively, with higher certainty in judgments about which candidate will win, particularly for the leading candidate (*top*) and more extreme judgments of anticipated vote share (*bottom*), even when accompanied by vote share projections ("both" condition). Participants are less accurate when attempting to judge the likelihood of winning (*middle*) than vote share (*top*). Plots on the right show differences when vote share is fixed at 55% (.87 probability). Lines fit using LOESS in plots on the left; results based on OLS regression in plots on the right, 95% confidence bands/intervals shown. Color version available as an online enhancement.

Reported uncertainty and public trust

The effects of communicating uncertainty on public trust in facts and numbers

Anne Marthe van der Bles[a,b,c,1], Sander van der Linden[a,b,d,1], Alexandra L. J. Freeman[a,b], and David J. Spiegelhalter[a,b]

[a]Winton Centre for Risk and Evidence Communication, University of Cambridge, Cambridge CB3 0WA, United Kingdom; [b]Department of Pure Mathematics and Mathematical Statistics, University of Cambridge, Cambridge CB3 0WA, United Kingdom; [c]Department of Social Psychology, University of Groningen, 19712 TS Groningen, The Netherlands; and [d]Cambridge Social Decision-Making Lab, Department of Psychology, University of Cambridge, Cambridge CB2 3RQ, United Kingdom

Edited by Arild Underdal, University of Oslo, Oslo, Norway, and approved February 20, 2020 (received for review August 7, 2019)

Uncertainty is inherent to our knowledge about the state of the world yet often not communicated alongside scientific facts and numbers. In the "posttruth" era where facts are increasingly contested, a common assumption is that communicating uncertainty will reduce public trust. However, a lack of systematic research makes it difficult to evaluate such claims. We conducted five experiments—including one preregistered replication with a national sample and one field experiment on the *BBC News* website (total *n* = 5,780)—to examine whether communicating epistemic uncertainty about facts across different topics (e.g., global warming, immigration), formats (verbal vs. numeric), and magnitudes (high vs. low) influences public trust. Results show that whereas people do perceive greater uncertainty when it is communicated, we observed only a small decrease in trust in numbers and trustworthiness of the source, and mostly for verbal uncertainty communication. These results could help reassure all communicators of facts and science that they can be more open and transparent about the limits of human knowledge.

communication | uncertainty | trust | posttruth | contested

the general sense of honesty evoked [by uncertainty] . . . this did not seem to offset concerns about the agency's competence" (p. 491). In fact, partly for these reasons, Fischhoff (1) notes that scientists may be reluctant to discuss the uncertainties of their work. This hesitation spans across domains: For example, journalists find it difficult to communicate scientific uncertainty and regularly choose to ignore it altogether (10, 11). Physicians are reluctant to communicate uncertainty about evidence to patients (12), fearing that the complexity of uncertainty may overwhelm and confuse patients (13, 14). Osman et al. (15) even go as far as to argue explicitly that "the drive to increase transparency on uncertainty of the scientific process specifically does more harm than good" (p. 131).

At the same time, many organizations that produce and communicate evidence to the public, such as the European Food Safety Authority, have committed themselves to openness and transparency about their (scientific) work, which includes communicating uncertainties around evidence (16–19). These attempts have not gone without criticism and discussion about the potential impacts on public opinion (15, 20). What exactly do we know about the effects of communicating uncertainty around

Table 1. Overview of the conditions and manipulation texts of experiment 3 and 4

Format	Experiment 3
Control (no uncertainty)	"Official figures from the first quarter of 2018 show that UK unemployment fell by 116,000 compared with the same period last year. [. . .]"
Numerical range with point estimate	. . .by 116,000 (range between 17,000 and 215,000). . .
Numerical range without point estimate	. . .by between 17,000 and 215,000. . .
Numerical point estimate ±2 SEs	. . .by 116,000 (±99,000). . .
Verbal explicit uncertainty statement	. . .by 116,000 compared with the same period last year, although there is some uncertainty around this figure: It could be somewhat higher or lower. [. . .]
Verbal implicit uncertainty statement	. . .by 116,000 compared with the same period last year, although there is a range around this figure: could be somewhat higher or lower. [. . .]
Verbal uncertainty word	. . .by an estimated 116,000. . .
Mixed numerical and verbal phrase	. . .by an estimated 116,000 (±99,000). . .

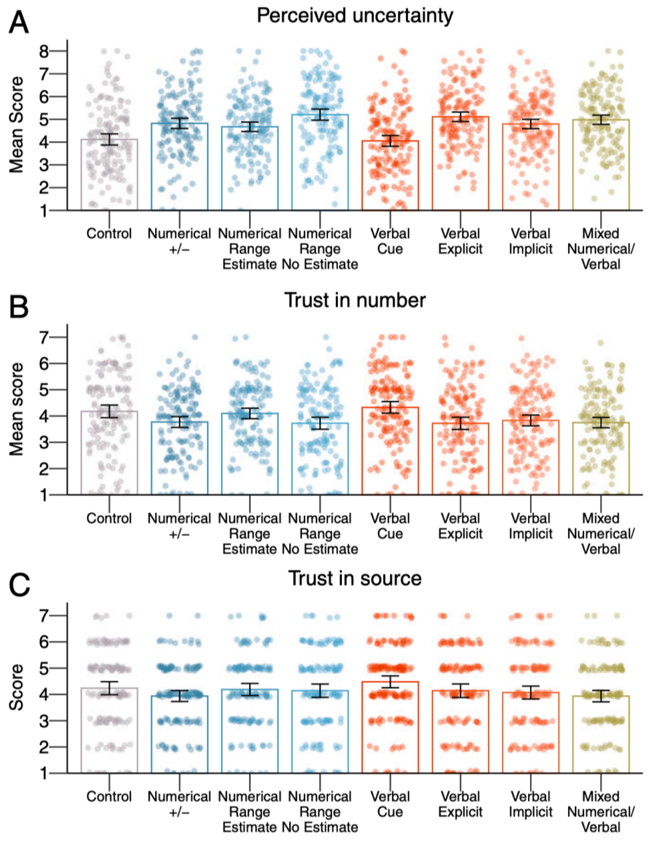
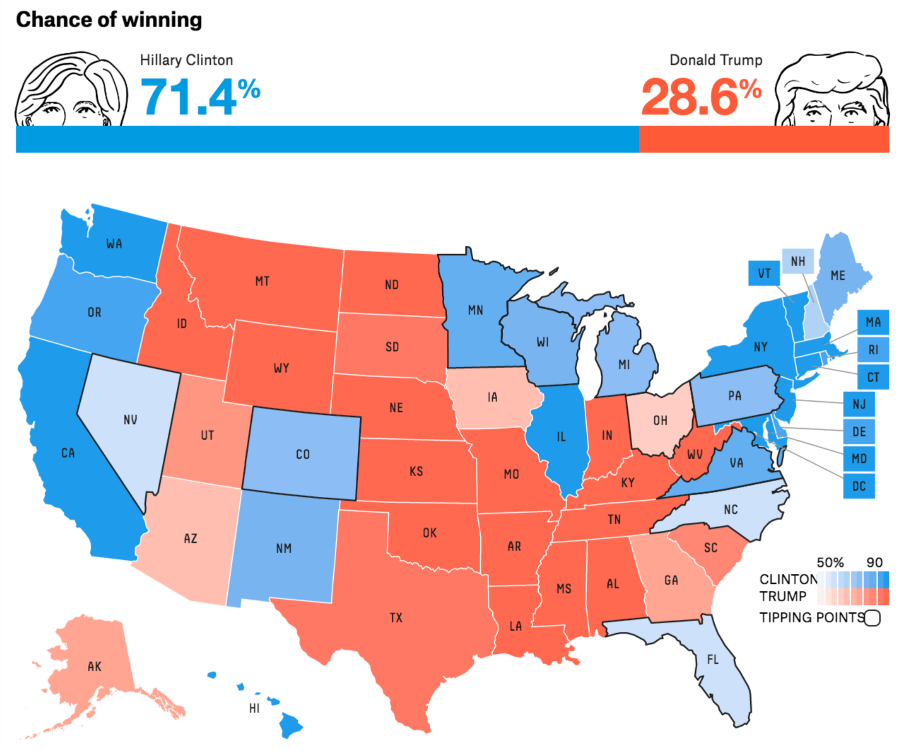


Fig. 3. The results of experiment 3: Means per condition for perceived uncertainty (*A*), trust in numbers (*B*), and trust in the source (*C*). The error bars represent 95% CIs around the means, and jitter represents the distribution of the underlying data.

`Source` Van der Bles et al. 2020, *PNAS*

But also...



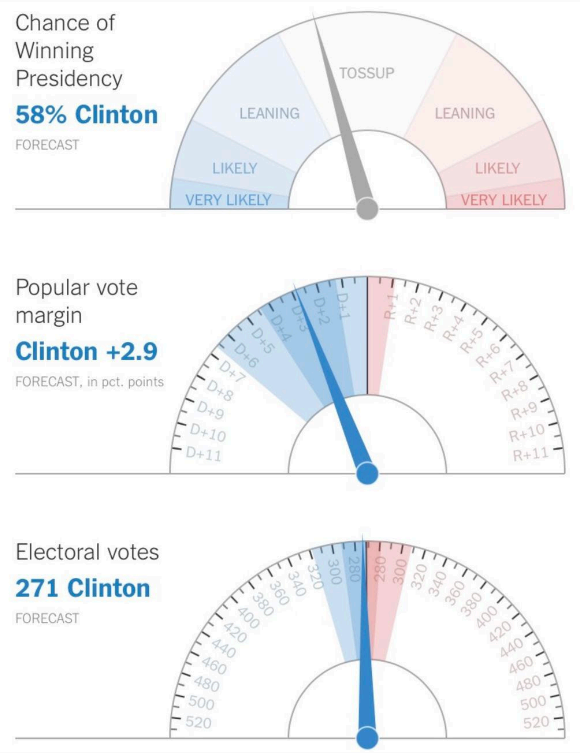
Electoral votes

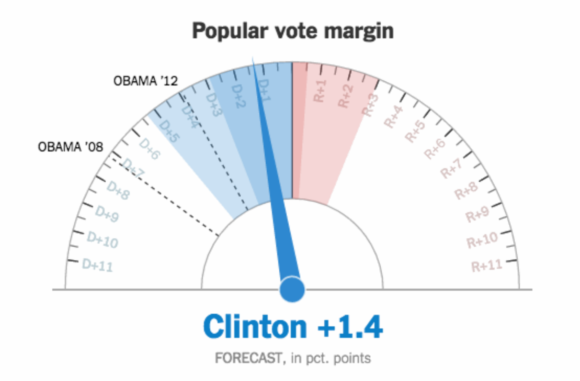
Hillary Clinton	302.2
Donald Trump	235.0
Evan McMullin	0.8
Gary Johnson	0.0

Popular vote

Hillary Clinton	48.5%
Donald Trump	44.9%
Gary Johnson	5.0%
Other	1.6%

Source FiveThirtyEight.com, 2016





Discussion

Class discussion

Questions for discussion

1. Skills for news consumption

Which skills matter most going forward — for audiences, journalists, or both?

2. Evidence for data journalism practice

Which scientific evidence is needed? Ideas for measures, designs, and outcomes that matter?

3. Closing the research–practice gap

How to bridge slow research and fast-paced newsrooms — and what role can journalists play in shaping the research agenda?

Takeaways: implications for data journalism practice

Communicating numbers and visuals

Most readers struggle with percentages and ratios — don't take anything stats-related for granted (Cokely et al. 2012)

Win probability ≠ vote share — readers systematically confuse the two, inflating perceived certainty and distorting the meaning of probabilistic forecasts (Westwood et al. 2020)

Charts can boost credibility — but modestly — adding data visualizations increases perceived persuasiveness and source credibility, yet effects are small and depend on prior beliefs (Link et al. 2021; Yang et al. 2023)

Communicating uncertainty

Reporting uncertainty does not erode trust — contrary to common editorial instinct, acknowledging data uncertainty leaves public trust in statements largely intact (van der Bles et al. 2020)

Uncertainty has multiple levels — source, measurement, model, and scientific consensus; conflating them misleads audiences (van der Bles et al. 2019)

Suppressing uncertainty is itself a form of misinformation — and one with a credibility cost when forecasts fail, as 2016 illustrated

Further reading